Application Layer Definitions

2.1 Principles of Network Applications

Application Architecture

- **Client-Server Architecture:** A model where an always-on host (the server) services requests from many other hosts (clients). Clients do not communicate directly with each other.
- **Peer-to-Peer (P2P) Architecture:** A model with no dedicated server. Hosts (peers) communicate directly with each other, acting as both clients and servers. Known for its *self-scalability*.

Processes & Communication

- **Process:** A program running on a host. Communicating processes on different hosts are the basis of a network application.
- **Socket:** The Application Programming Interface (API) that acts as a "door" between an application process and the transport layer. An application sends and receives all network messages through a socket.
- **IP Address:** A 32-bit (IPv4) numerical identifier for a host on the internet.
- **Port Number:** A 16-bit number used with an IP address to identify a specific process on a host.

Transport Layer Services

- **TCP (Transmission Control Protocol):** A reliable, connection-oriented transport protocol. It provides in-order data delivery, flow control, and congestion control.
- **UDP (User Datagram Protocol):** An unreliable, connectionless transport protocol. It offers minimal, "best-effort" service with no guarantees.
- **TLS (Transport Layer Security):** An application-layer enhancement that runs on top of TCP to provide encryption, data integrity, and authentication for secure communication (e.g., HTTPS).

2.2 Web and HTTP

Core Web Concepts

- **Web Page:** A document consisting of a base HTML file and several referenced *objects* (images, scripts, etc.).
- **URL (Uniform Resource Locator):** An address specifying the hostname and path to a web object.
- **HTTP (Hypertext Transfer Protocol):** The application-layer protocol for the web. It is a *pull protocol* (clients pull data from servers) and is **stateless**.
- **Stateless Protocol:** A protocol where the server maintains no information about past client requests.

HTTP Connections

- **Non-persistent HTTP:** A version of HTTP where a separate TCP connection is used for every object. Inefficient due to high overhead (2 RTTs per object).
- **Persistent HTTP:** The default in modern HTTP. Multiple objects are transferred over a single TCP connection, reducing overhead significantly.
- **RTT (Round-Trip Time):** The time it takes for a small packet to travel from a client to a server and back again.

Cookies & State

- **Cookies:** A mechanism used by websites to track user state. The server sends a unique ID to the client, which the client sends back in subsequent requests.
- **First-Party Cookie:** A cookie associated with the domain the user is directly visiting.
- **Third-Party Cookie:** A cookie associated with a domain different from the one the user is visiting, typically used by ad networks to track users across multiple websites.

Performance Optimization

- **Web Cache (Proxy Server):** A network entity that stores copies of recently requested web objects to satisfy future requests locally, reducing response time and network traffic.
- **Conditional GET:** An HTTP mechanism that allows a cache to verify if its stored object is up-to-date. If not, the server sends a small '304 Not Modified' response instead of the full object.
- **HTTP/2:** An evolution of HTTP that reduces delay by mitigating Head-of-Line (HOL) blocking. It divides objects into frames and interleaves them on a single TCP connection.
- **HTTP/3:** The newest version, which uses QUIC (over UDP) to eliminate transport-layer HOL blocking.

2.3 Email Protocols

- **User Agent (UA):** The client software used to compose, read, and manage email (e.g., Outlook, Apple Mail).
- **Mail Server:** An always-on host that stores incoming emails in user **mailboxes** and outgoing emails in a **message queue**.
- **SMTP (Simple Mail Transfer Protocol):** The application-layer protocol for **pushing** email from a client to a server, and from one server to another.
- **Mail Access Protocol:** A protocol used by a user agent to **pull** email from a mail server.
  - **IMAP (Internet Mail Access Protocol):** A modern access protocol where messages are managed and stored on the server, allowing for a synchronized state across multiple clients.
  - **POP (Post Office Protocol):** An older protocol where messages are typically downloaded to a single client and removed from the server.

2.4 The Domain Name System (DNS)

- **DNS:** A distributed, hierarchical database implemented as an application-layer protocol that translates human-readable hostnames into IP addresses.
- **DNS Hierarchy:** The structure of DNS servers.
  - **Root DNS Servers:** Top of the hierarchy; they direct queries to the appropriate TLD servers.
  - **Top-Level Domain (TLD) Servers:** Responsible for domains like '.com', '.org', '.edu', and country codes.
  - **Authoritative DNS Servers:** An organization's server that holds the definitive hostname-to-IP mappings for its domain.
- **Local DNS Server:** A server (not in the hierarchy) that acts as a proxy, forwarding DNS queries from local hosts into the DNS hierarchy.
- **Iterated Query:** A DNS query style where the local server asks each server in the hierarchy directly, one after another.

- **Recursive Query:** A DNS query style where the burden of resolution is passed up the hierarchy.
- **DNS Caching:** The practice of storing DNS query results locally to speed up future requests for the same hostname. The duration of caching is controlled by a **Time-to-Live (TTL)** value.
- **Resource Record (RR):** An entry in the DNS database. Key types include **A** (address), **NS** (name server), **CNAME** (alias), and **MX** (mail server).

2.5 & 2.6 P2P, Video, and CDNs

- **BitTorrent:** A popular P2P protocol for file distribution where a file is broken into **chunks**.
  - **Tracker:** A server that tracks the peers participating in a **torrent**.
  - **Tit-for-Tat:** The incentive mechanism where a peer uploads fastest to the peers that upload fastest to it.
- **Jitter:** The variability of packet delay over time. A key challenge in video streaming.
- **Playout Buffer:** A client-side buffer used in streaming applications to absorb jitter and ensure smooth, continuous playout of video.
- **DASH (Dynamic Adaptive Streaming over HTTP):** The dominant video streaming standard. The client detects its current bandwidth and requests video chunks of a corresponding quality level from the server.
- **Manifest File:** A file provided by a DASH server that contains the URLs for all the different quality versions of all video chunks.
- **CDN (Content Distribution Network):** A geographically distributed network of servers that stores copies of content (like videos) to bring it closer to end-users, improving performance and scalability.